RVA Specialist Coding Assessment

Roche PD Data Science & Analytics

Introduction

Welcome to the R Shiny Clinical Data Assessment. This exercise evaluates your proficiency in data manipulation with `{tidyverse}`, clinical reporting with `{gtsummary}`, and interactive application development using **R Shiny**.

This assessment includes practical coding exercises focused on clinical trial reporting in the pharmaceutical industry. It is designed to evaluate your technical expertise and critical thinking. Candidates are allowed to use AI-enabled coding assistants.

Assessment Format

This assessment consists of three core questions utilizing the `ADSL` and `ADAE` datasets from the `pharmaverseadam` package.

Submission Guidelines

1. **GitHub Repository:**
   - Create a public repository for your solutions
   - Repository should be well-organized with clear folder structure (one folder for each question)
   - Include a `README.md` explaining the repo structure and the contents of each folder that could be helpful for the reviewer
2. **Code Files:**
   - Each question should have a dedicated folder and its own file (e.g., `question_1.R`, `question_2.R`, etc.)
3. **Video Explanation:**
   - Record a brief video walkthrough of your repository
   - Explain your approach, design decisions, and key challenges overcome
   - Discuss what you learned from the exercise

Timeline

- **Submission Deadline:** 1 week
- **Submission Format:** GitHub repository link and video walkthrough submitted as part of the Phenom interview
- **Interview Scheduling:** Selected candidates will be contacted for follow-up discussion

Evaluation Criteria

Your submission will be evaluated on:

- **Code Quality:** Clarity, efficiency, and adherence to R best practices
- **Correctness:** Does the code produce the expected clinical outputs?
- **Documentation:** Are functions and code well-documented?

- **Problem-Solving:** How efficient and creative is your use of the provided tools?
- **Communication:** Can you explain your approach clearly?

Posit Cloud

Candidates can use a Posit Cloud free plan or any other plan as mentioned in the Posit website (https://posit.cloud/plans) to work on R related questions. Once logged in, create a new workspace to clone your GitHub repository to get started.

R Version

R 4.2.0 and above.

Required R Packages

The necessary packages, along with all dependencies, can be installed from CRAN using the provided command.

```
install.packages(c("pharmaverseadam", "tidyverse", "gtsummary", "ggplot2", "shiny"))
```

Question 1: TEAE Summary Table

Create a regulatory-compliant summary table of **Treatment-Emergent Adverse Events (TEAEs)**.

- **Input Data:** `pharmaverseadam::adae` and `pharmaverseadam::adsl`.
- **Requirements:**
  - Treatment-emergent AE records will have `TRTEMFL == "Y"`.
  - **Rows:** System Organ Class (`AESOC`) and Preferred Term (`AEDECOD`).
  - **Columns:** Treatment groups (`ACTARM`).
  - **Cell Values:** Subject count (n) and percentage (%).
  - **Denominator:** Use the total number of subjects in the study from ADSL.
  - **Total:** Include a summary row at the top for all TEAEs.
- **Tooling:** Use `{gtsummary}`.
- **Sample Output:**

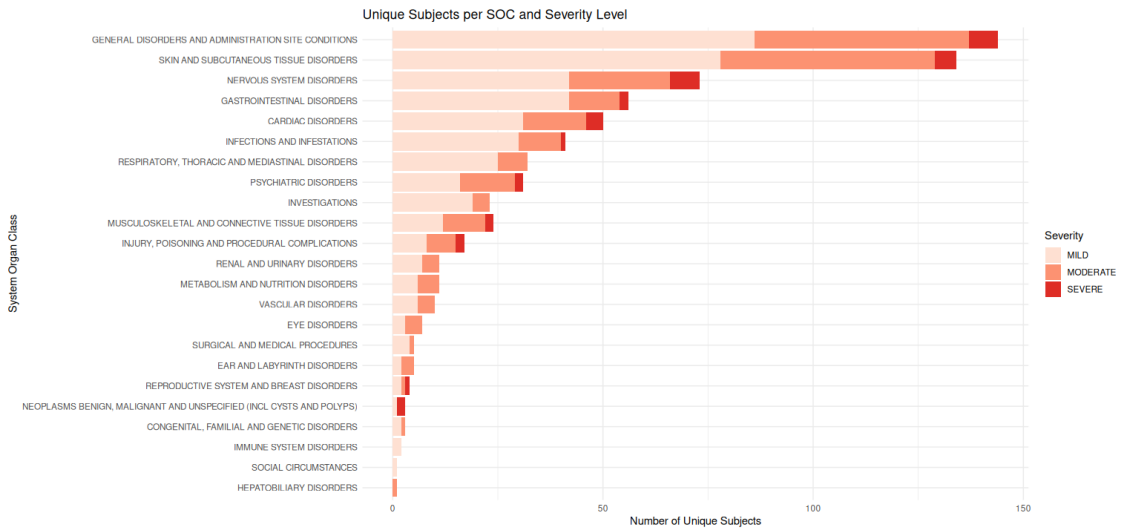
System Organ Class / Preferred Term	Placebo N = 86[1]	Xanomeline High Dose N = 72[1]	Xanomeline Low Dose N = 96[1]
Treatment Emergent Adverse Events	65 (76%)	68 (94%)	84 (88%)
CARDIAC DISORDERS	12 (14%)	14 (19%)	14 (15%)
ATRIAL FIBRILLATION	1 (1.2%)	2 (2.8%)	2 (2.1%)
ATRIAL FLUTTER	0 (0%)	1 (1.4%)	1 (1.0%)
ATRIAL HYPERTROPHY	1 (1.2%)	0 (0%)	0 (0%)
ATRIOVENTRICULAR BLOCK FIRST DEGREE	1 (1.2%)	0 (0%)	1 (1.0%)
ATRIOVENTRICULAR BLOCK SECOND DEGREE	1 (1.2%)	0 (0%)	0 (0%)
BRADYCARDIA	1 (1.2%)	0 (0%)	0 (0%)

- **Output Format:** HTML file

Question 2: AE Severity Visualization

Develop a publication-quality bar chart visualizing the distribution of adverse events.

- **Input Data:** `pharmaverseadam::adae`.
- **Requirements:**
  - **X-axis:** Count of unique subjects per System Organ Class per Severity. Ensure each subject subject is counted at most once per severity level within each SOC.
  - **Y-axis:** System Organ Class term (`AESOC`).
  - **Color/Fill:** Colored by AE Severity/Intensity (`AESEV`).
  - **Ordering:** Bars must be stacked and ordered by increasing frequency of total subjects per SOC.
- **Tooling:** Use `{ggplot2}`.
- **Sample Output:**



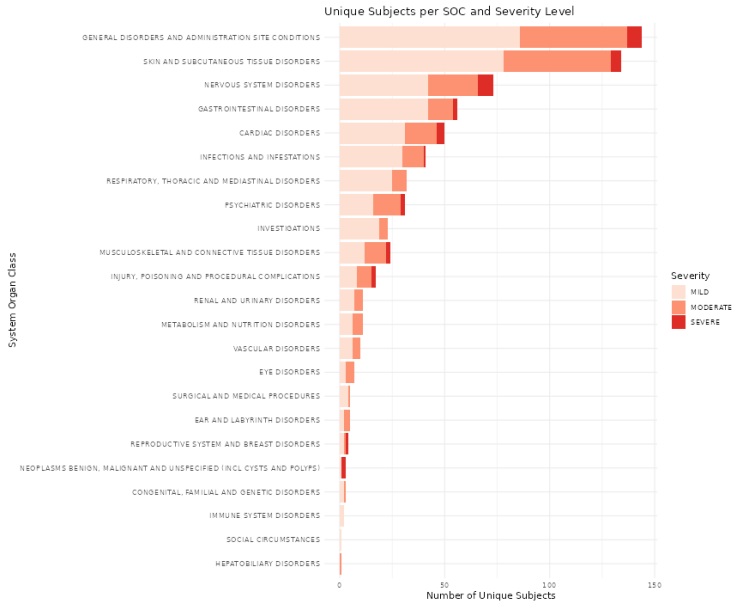
- **Output Format:** PNG file

Question 3: Interactive R Shiny Application

Integrate the visualization from Question 2 into an interactive dashboard.

- **Requirements:**
  - Display the bar chart from Question 2 within a Shiny UI.
  - **Filter:** Add a user input (e.g., `selectInput` or `checkboxGroupInput`) to filter the chart by Treatment Arm (`ACTARM`).
  - **Reactivity:** The plot must update dynamically based on the selected filter.
- **Tooling:** Use the `{shiny}` framework.
- **Sample Output:**

AE Summary Interactive Dashboard

Select Treatment Arm(s):

- [x] Placebo
- [x] Xanomeline High Dose
- [x] Xanomeline Low Dose